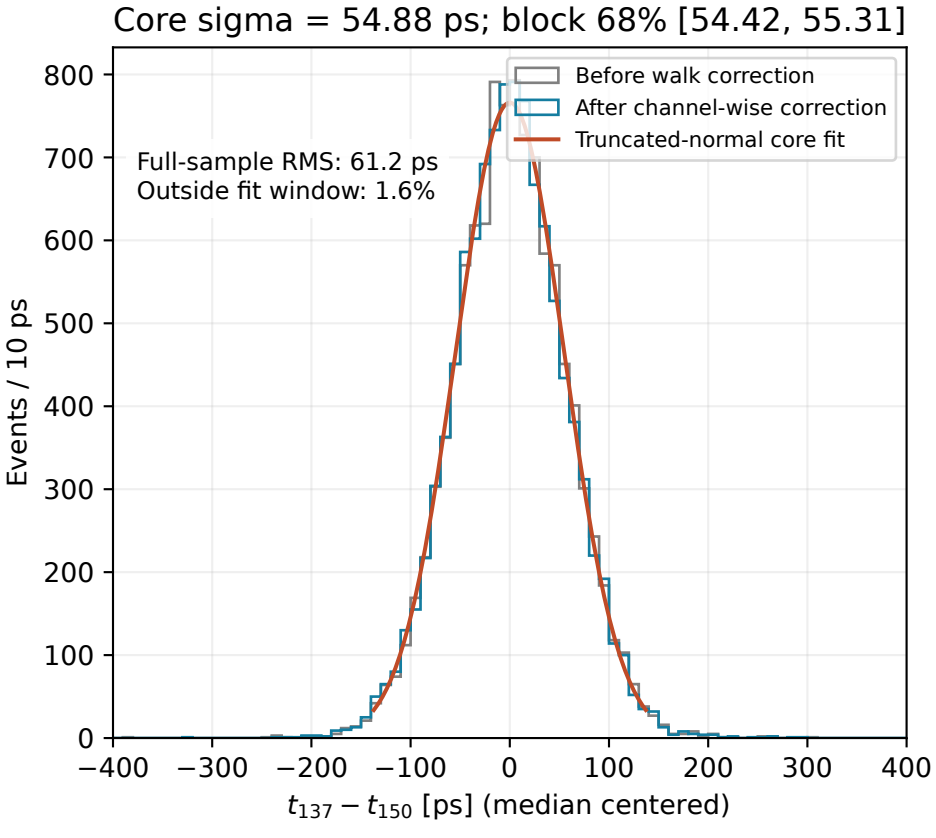
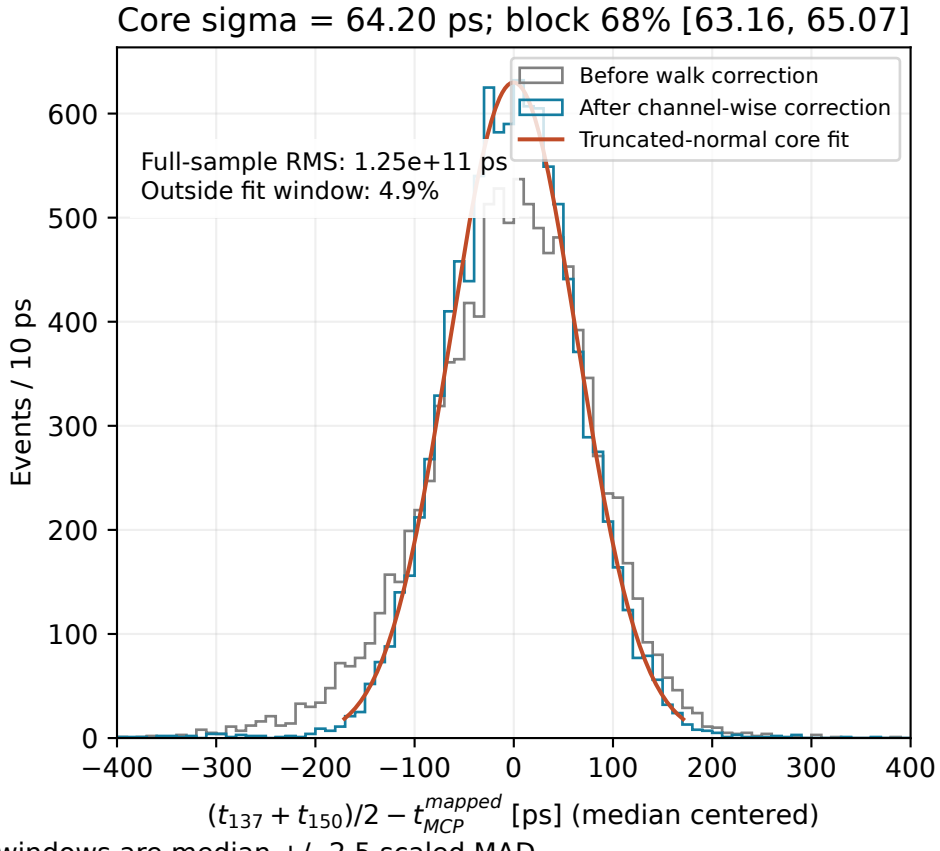
CMS Preliminary Phase 2

Run 4399 | Bar 8 | 10,577 common selected events

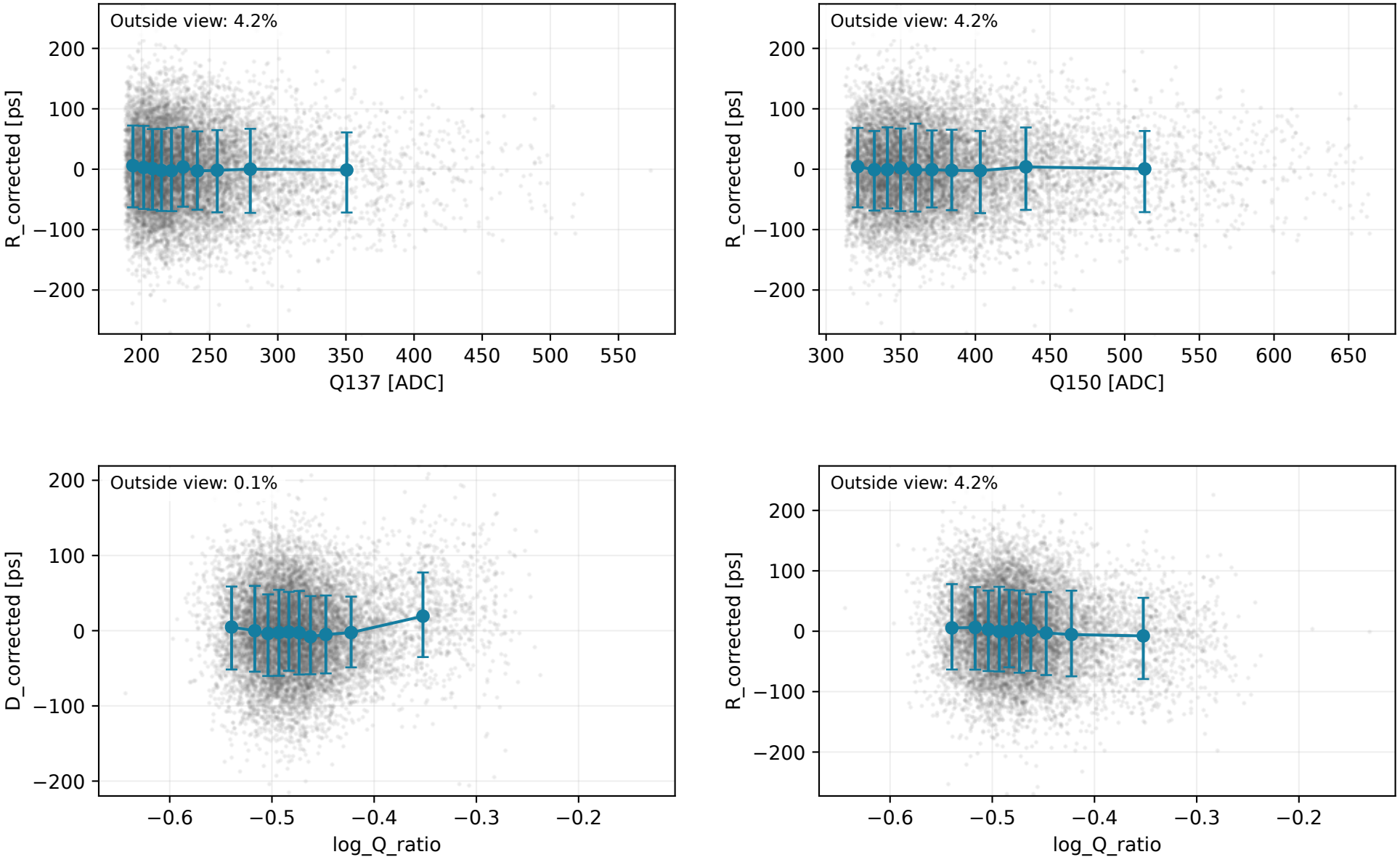




No timing-tail selection. Each distribution is centered only for display; fit windows are median +/- 2.5 scaled MAD.
Bootstrap intervals are conditional on frozen clock and walk calibrations. Display range does not restrict the reported RMS.

CMS Preliminary Phase 2

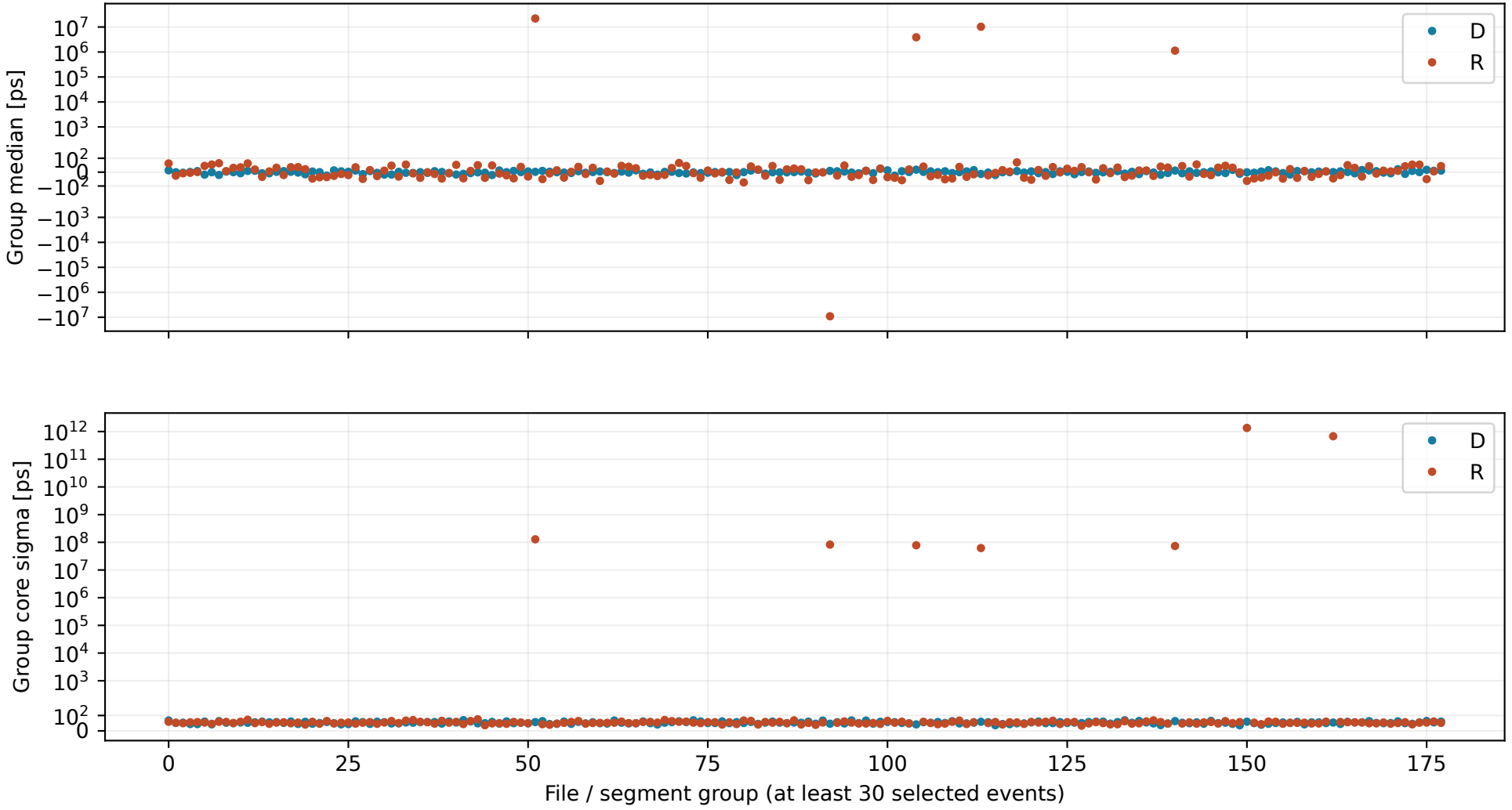
Charge and charge-asymmetry dependence



All events enter the summaries; view is median +/- 4 scaled MAD (outside-view fractions shown). Ten equal-occupancy bins show median and central 68% event spread, not SEM. Charge asymmetry is a proxy, not independent tracking.

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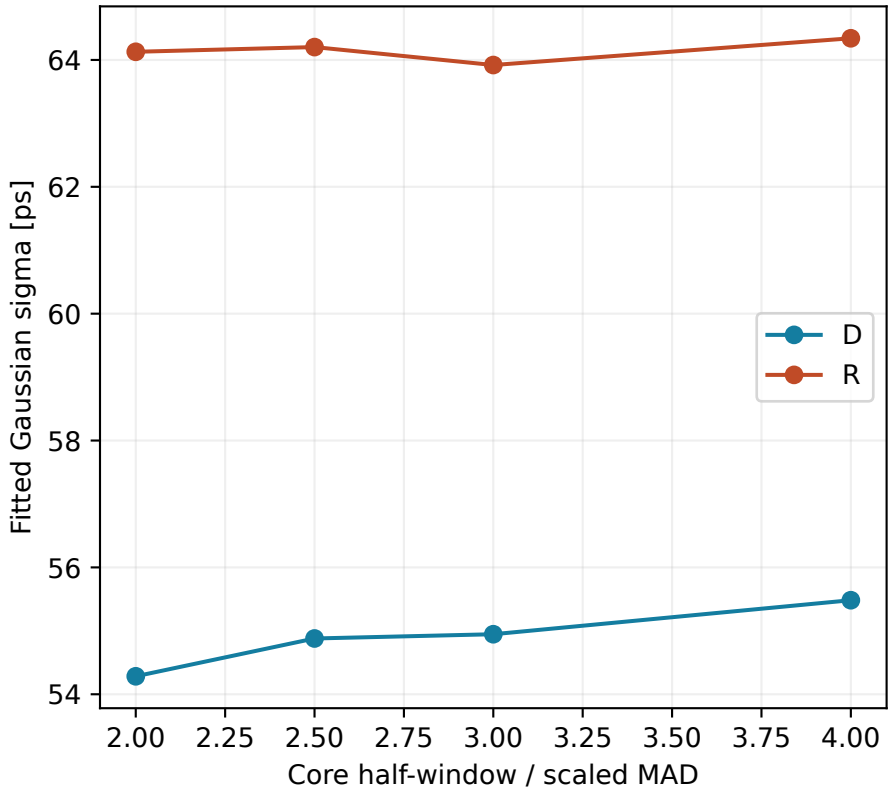
Stability across file / scope-segment groups

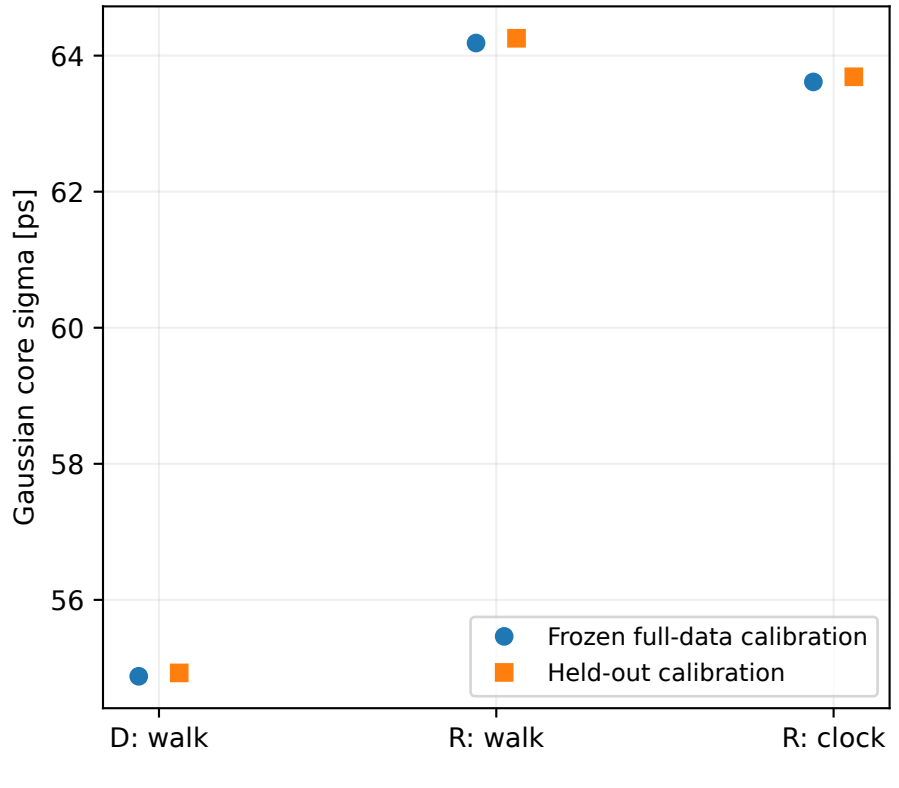


D: end difference. R: bar-average minus mapped MCP. No per-group recentering is applied to the headline distributions.
Group widths are diagnostic point estimates; exact identifiers and counts are saved in segment_stability.csv.

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Fit-window sensitivity and held-out calibration checks

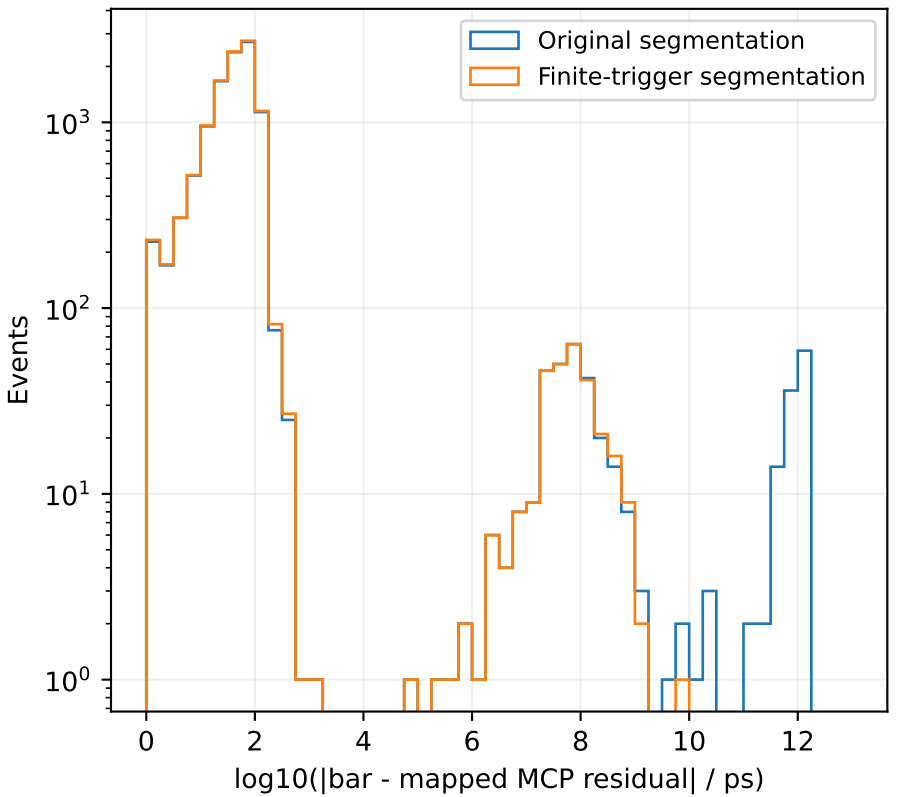


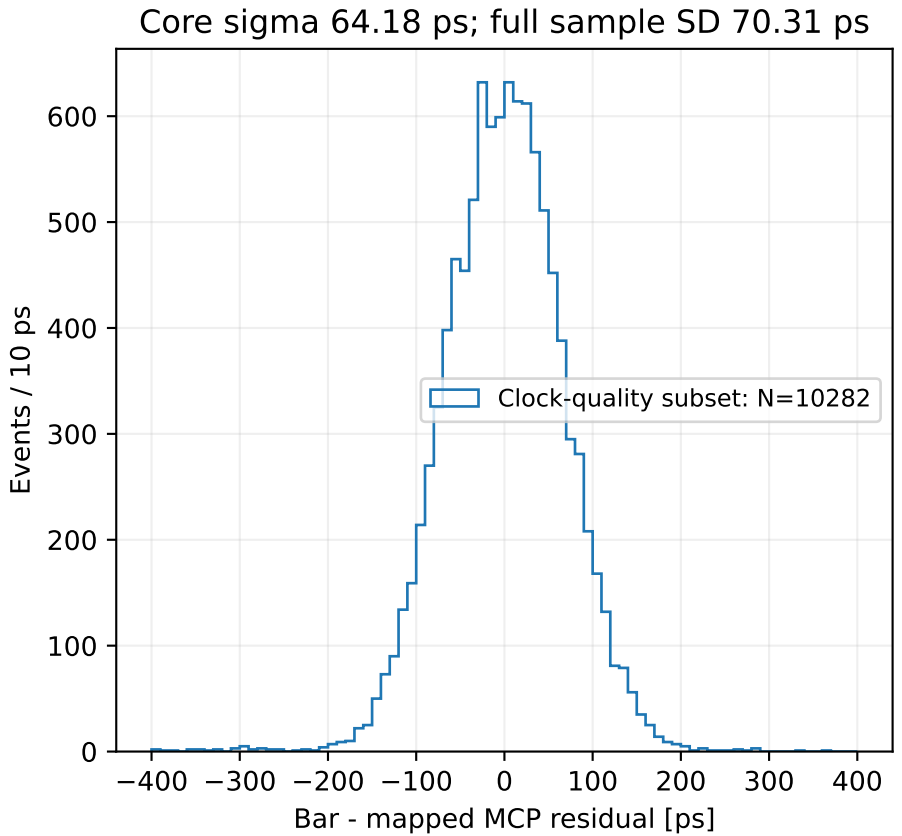


Walk: hold out each scope-segment ordinal across files, keeping the original clock mapping.
Clock: alternate trigger events within each segment; no extrapolation. Width pairs use identical eligible events.
These separate diagnostics are not a fully independent joint clock-and-walk validation.

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Clock validation: failures hidden by a core-only plot

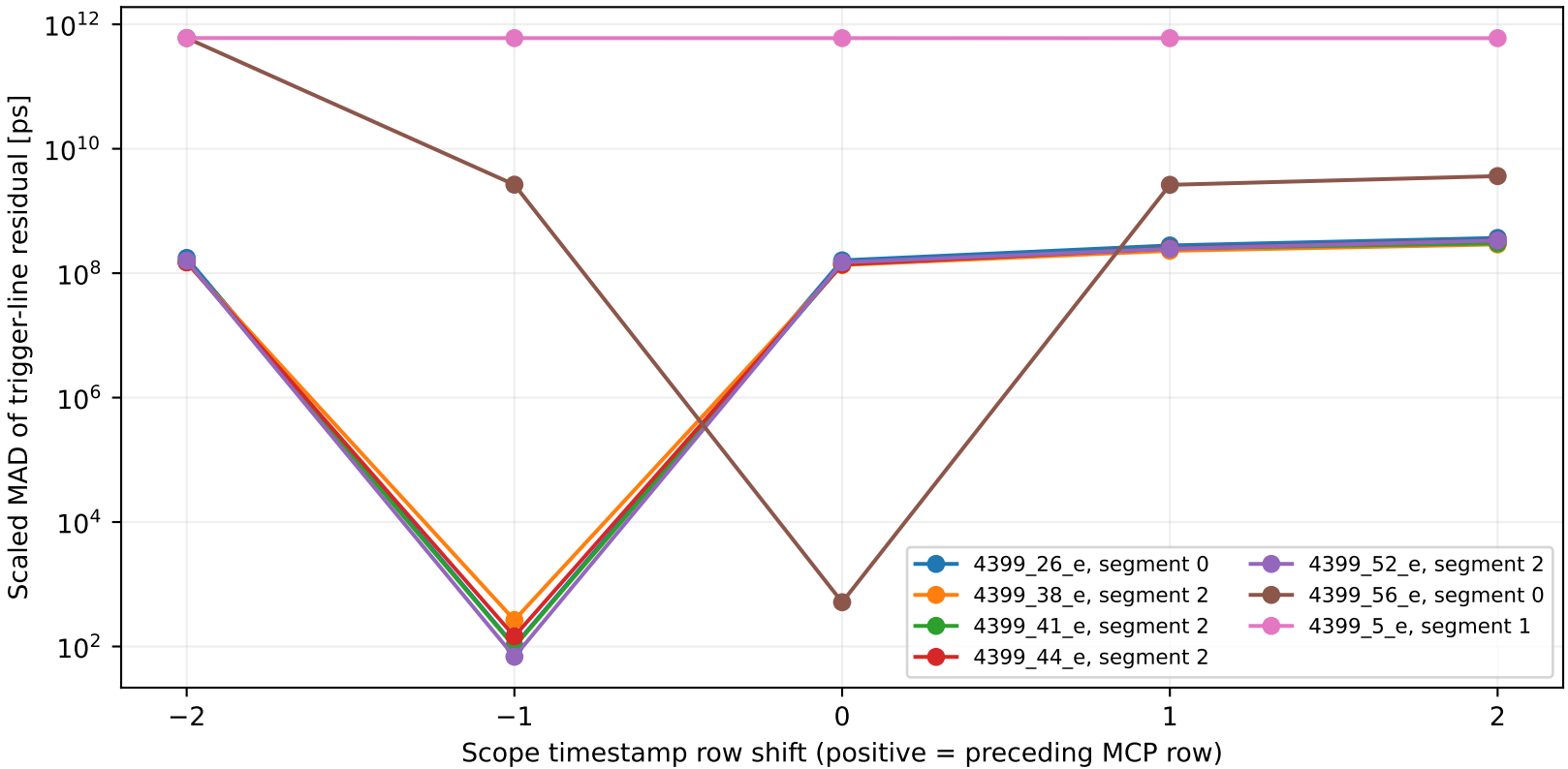




Local diagnostic only: segment finite scope triggers across NaN gaps, then reject entire clock groups with line-residual scaled MAD >= 1 microsecond. This uses clock signals, not the SiPM timing residual.
Frozen walk curves are unchanged. No production fix or final resolution claim is made.

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Diagnostic of a possible one-event matching offset



Row shifts are a diagnostic, not an applied correction. A large improvement at a one-row shift warrants an audit of event association upstream of clock calibration. This does not establish a safe remapping rule.